DAILY SPORTS APP

SLACK MANAGEMENT & PDM ANALYSIS REPORT

Client: Foreign Mobile App Development Company
Authors: Soroush Sahraei & Mahdis Mirzaei (Group 7)
Timeline: April 1, 2026 – January 31, 2027 (10-month window)

1. INITIAL PROJECT PLAN & MANUAL CPM

The objective of this project is to develop and launch a daily sports training and management application. The project consists of 9 distinct tasks mapped using the Precedence Diagramming Method (PDM). All dependencies are defined as Finish-to-Start (FS) with zero lag.

#	Task Name	Duration	Dependencies
1	Software Development	3 months	None
2	Software Integration	1 month	Task 1
3	UI/UX Design	2 months	None
4	Cross-Platform α Test	1 month	Tasks 2, 3
5	Implementing α Feedback	1 month	Task 4
6	Selected-User β Test	1 month	Task 5
7	Developing Final Package	2 months	Task 6
8	Application Marketing	1 month	Task 6
9	Final Launch (Milestone)	0 months	Tasks 7, 8

Manual Calculation Methodology (CPM):

- **Forward Pass:** Early Finish is calculated as $EF_j = ES_j + D_j$, and Early Start for dependent tasks is $ES_j = \max\{EF_i\}$.
- **Backward Pass:** Late Start is calculated as $LS_i = LF_i - D_i$, and Late Finish for preceding tasks is $LF_i = \min\{LS_j\}$.
- **Slack Formula:** Total Slack (TS = LS - ES) and Free Slack (FS = $\min\{ES_j\} - EF_i$).

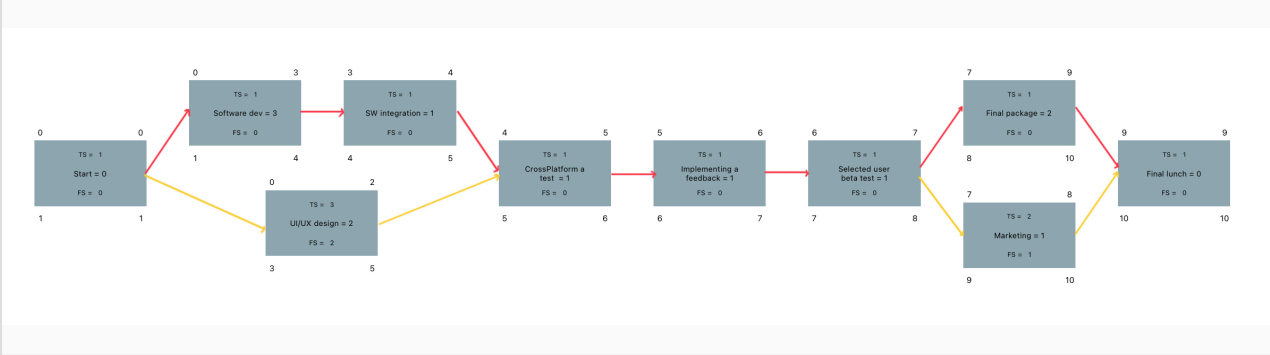


Figure 1: Manual PDM Network Diagram

2. SOFTWARE IMPLEMENTATION & VALIDATION

To validate the manual calculations, the project was imported into Microsoft Project. A custom 24/7 calendar was created, and MS Project's schedule settings were adjusted so that 1 month equals exactly 30 days to align with manual calculations.

ID	Task	Critical?
1	Software Development	Yes
2	Software Integration	Yes
3	UI/UX Design	No
4	Cross-Platform α Test	Yes
5	Implementing α Feedback	Yes
6	Selected-User β Test	Yes
7	Developing Final Package	Yes
8	Application Marketing	No
9	Final Launch	Yes

Comparison: Manual vs. Software Calculations

The MS Project results match the manual CPM calculations exactly. Both methods use the same forward/backward pass logic. Differences can arise from calendar settings (e.g., weekends off vs. continuous work) or deadline constraints, but a custom 24/7 calendar was applied here.



Figure 2: MS Project Detail Gantt showing Calculated Values and Slack

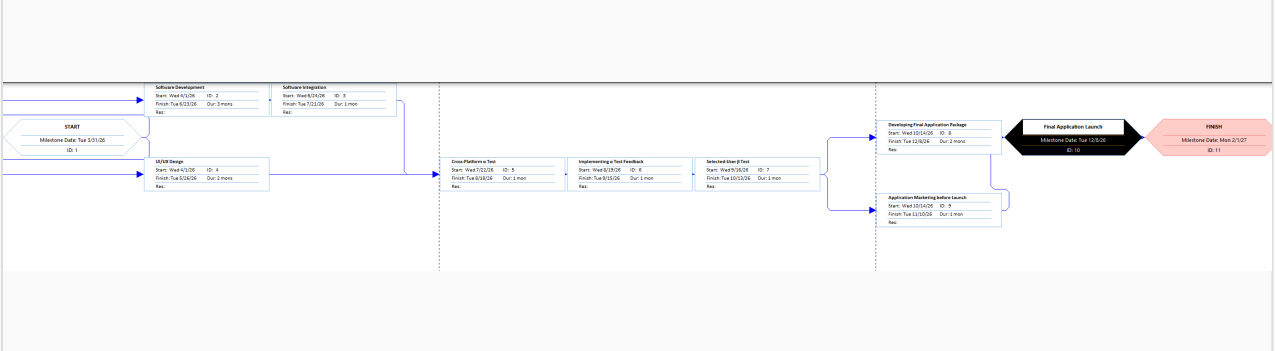


Figure 3: MS Project PDM Network Diagram highlighting the Critical Path

3. INTERPRETATION OF SLACK CAUSES

Primary Development Path

Slack: 1.95 Months (MS Project). The tasks on the main development branch share this slack. This exists purely because the client provided a 10-month delivery window for 9 months of planned work. This is shared project slack, meaning if Task 1 is delayed by a month, the team will consume this buffer, leaving no room for errors in later testing phases.

UI/UX Design (Task 3)

Free Slack: 2 Months. This task runs parallel to a longer software development branch. Tasks 1 and 2 require 4 months of work, while Task 3 only requires 2 months. Therefore, the design team has 2 months of Free Slack to start late without affecting the Alpha Test at all.

Application Marketing (Task 8)

Free Slack: 1 Month. Marketing runs parallel to the Final Package development (Task 7). Task 7 takes 2 months, whereas Task 8 only takes 1 month. The marketing team can use this 1 month of independent slack without delaying the Final Launch.

4. EXECUTIVE SUMMARY

The total length of the project's primary path (Tasks 1 > 2 > 4 > 5 > 6 > 7 > 9) is **9 months**.

Because the client has provided a 10-month delivery window (April 2026 to January 2027), the project is highly viable and carries a natural project-level buffer. While MS Project shows 1.95 months of Total Slack due to its internal calendar, realistically, the team has about 1 calendar month of safety. We recommend keeping the schedule tight and saving this buffer for the Alpha and Beta testing phases, as user feedback often requires unexpected time to fix.

REFERENCES

[1] Project Management Institute. *PMBOK Guide* – 7th Edition.

[2] Kerzner, H. *Project Management: A Systems Approach*.

[3] Microsoft Project Built-in Help Documentation.